

Lecture 07 - Exercises

These exercises accompany Lecture 07, “Sampling Distributions and Confidence Intervals.” They practice the logic of inference from samples: sampling design, sampling fluctuation, standard error, the sampling distribution of the mean, confidence intervals, and careful interpretation.

The exercises are separated into two parts. The first part should be done by hand. This is important because manual work helps you understand the mechanics: what the parameter is, what the estimator is, how the standard error is built, how the critical value affects the interval, and what the interval actually means. Only after that does it become useful and convenient to move to larger datasets and use Python.

The second part uses Python. These exercises apply the same ideas to return and credit datasets. Python is used to compute standard errors, build confidence intervals, compare confidence levels, and write careful precision statements.

Throughout, we use normal critical values (1.645, 1.96, 2.58) as a working approximation, exactly as in the lecture. This is a deliberate simplification suitable for an introductory course and for the large samples used here.

Solutions are not included here. They will be provided separately: the by-hand solutions in Mark-down and the Python solutions in a notebook.

Part A - By-Hand Practice

Exercise 1 - Sampling and Inference

A manufacturer produces climbing ropes. Testing a rope’s strength requires pulling it until it breaks.

1. What is the population?
2. What is a possible sample?
3. What parameter might the company want to estimate?
4. What estimator might be used?
5. Why is sampling necessary in this example?
6. What could go wrong if the sampled ropes are not representative?

Exercise 2 - Sampling Methods

For each situation, identify the sampling method as random sampling, systematic sampling, or quota sampling. If a situation does not fit any of these three cleanly, say so and name the method it most resembles (for example, convenience sampling). Then name one possible benefit and one possible risk.

1. A company selects every 50th transaction from a database.
2. A survey firm chooses respondents so the sample has the same age-group proportions as the population.
3. A quality-control team uses a random number generator to select 80 products from today’s production.
4. A retailer surveys the first 100 customers who enter one store on Monday morning.

Exercise 3 - Parameter, Estimator, and Estimate

A portfolio manager wants to estimate the long-run expected monthly return of a strategy. She observes 36 monthly returns and computes an average monthly return of 0.8%.

Identify:

1. the population or process;
2. the parameter;
3. the estimator;
4. the estimate;
5. one reason the estimate may differ from the parameter.

Exercise 4 - Sampling Fluctuation

Suppose four independent samples of 30 monthly returns from the same process give sample means:

Sample	Sample mean
1	-0.25%
2	0.30%
3	1.78%
4	0.09%

1. Why are the four sample means different?
2. Does the difference prove that the underlying process changed?
3. What is the name of the distribution formed by many such sample means?
4. Why does this idea matter for statistical inference?

Exercise 5 - Standard Deviation or Standard Error?

For each statement, say whether it refers mainly to the standard deviation or the standard error.

1. Customers differ a lot in their waiting times.
2. The estimated average waiting time is not very precise.
3. Daily returns are very volatile.
4. The sample mean return could change noticeably if we used another sample.
5. Individual delivery times are spread out.
6. The estimate of the average delivery time is based on a large sample and is therefore relatively stable.

Exercise 6 - Computing Standard Error

For each case, compute the standard error of the sample mean.

1. $\sigma = 1.2$, $n = 100$.
2. $\sigma = 5$, $n = 25$.
3. $\sigma = 5$, $n = 100$.
4. $\sigma = 5$, $n = 400$.

Then explain what happens to the standard error when sample size increases.

Exercise 7 - Sample Size and Precision

Two teams estimate average customer waiting time.

Team	Sample mean	Sample standard deviation	Sample size
A	4.0 minutes	2.0 minutes	25
B	4.0 minutes	2.0 minutes	100

1. Compute the estimated standard error for Team A.
2. Compute the estimated standard error for Team B.
3. Which estimate is more precise?
4. Why can the sample means be equal but the evidence not equally precise?

Exercise 8 - Central Limit Theorem

For each statement, say whether it is careful or incorrect. Rewrite if needed.

1. "If the sample size is large, the raw data become normally distributed."
2. "If the sample size is large, the sampling distribution of the sample mean is often approximately normal."
3. "The Central Limit Theorem means any sample of size 5 is enough."
4. "The normal approximation is more questionable when observations are strongly dependent or dominated by extreme values."
5. "The standard error is the standard deviation of the sampling distribution of the estimator."

Exercise 9 - Netflix Sampling Distribution

Suppose weekly Netflix viewing time in a population has mean 6 hours and standard deviation 1.2 hours. We repeatedly take samples of 100 households and compute the sample mean for each sample.

1. What is the mean of the sampling distribution of the sample mean?
2. Compute the standard error.
3. Interpret the standard error in words.
4. Why is the standard error smaller than the population standard deviation?

Exercise 10 - Standard Normal Critical Values

Use the common central normal critical values:

Confidence level	Critical value
90%	1.645
95%	1.96
99%	2.58

1. Which confidence level gives the narrowest interval, holding the standard error fixed?
2. Which confidence level gives the widest interval?
3. Why does higher confidence require a wider interval?
4. If the standard error is 0.20, compute the margin of error for each confidence level.

Exercise 11 - Central Interval for the Sample Mean (Known Parameters)

In this exercise the population parameters are treated as known, so nothing is being estimated. We are describing where sample means fall.

Suppose sample means are approximately normal with population mean 30, population standard deviation 5, and sample size 50.

1. Compute the standard error.
2. Construct the central interval that contains about 95% of sample means, centered on the known population mean, using the critical value 1.96.
3. Interpret the interval: across repeated samples of size 50, roughly what fraction of sample means fall inside it?
4. Because the population mean is known here, explain why this interval is a statement about the sampling distribution and is *not* a confidence interval.

Exercise 12 - Netflix Central Interval for the Sample Mean (Known Parameters)

Use the Netflix viewing example, again treating the parameters as known:

$$\mu = 6, \quad \sigma = 1.2, \quad n = 100.$$

1. Compute the standard error.
2. Compute the 95% margin using critical value 1.96.
3. Construct the central 95% interval for the sample mean, centered on the known population mean.
4. Interpret the interval in words (in terms of sample means across repeated samples).
5. Explain how this interval would differ if the population mean were *unknown* and had to be estimated from a single sample. What would the interval be centered on then?

Exercise 13 - Confidence Interval from Sample Data (Unknown Mean)

Now move to the realistic case: the population mean is unknown and must be estimated from the sample.

A fund has 64 monthly excess returns. The sample mean is 0.8% and the sample standard deviation is 4.0%.

1. Estimate the standard error.
2. Construct an approximate 95% confidence interval using 1.96.
3. Does the interval include zero?
4. Write a careful interpretation.
5. Explain why this interval is centered on the sample mean rather than on a known population mean, contrasting it with the intervals in Exercises 11 and 12.

Exercise 14 - Confidence Level and Interval Width

An estimate is 50 and the standard error is 2.

1. Construct a 90% interval.
2. Construct a 95% interval.
3. Construct a 99% interval.

4. Explain how the confidence level changes the interval width.

Exercise 15 - Benchmark Interpretation

A company wants average delivery time to be below 3 days. A sample gives:

Quantity	Value
Sample mean	2.8 days
Standard error	0.15 days

1. Construct an approximate 95% confidence interval using 1.96.
2. Is the sample mean below the benchmark?
3. Does the interval include the benchmark?
4. Write a careful conclusion about whether the evidence clearly supports average delivery time below 3 days.

Exercise 16 - Confidence Interval or Future-Outcome Range?

For each statement, say whether it is careful or incorrect. Rewrite if needed.

1. “The 95% confidence interval for average waiting time is [3.6, 4.4] minutes, so 95% of customers will wait between 3.6 and 4.4 minutes.”
2. “The interval [3.6, 4.4] minutes gives a plausible range for the unknown average waiting time, under the assumptions of the method.”
3. “The standard deviation describes how spread out individual observations are.”
4. “The standard error describes how much an estimate may vary from sample to sample.”
5. “A narrow confidence interval always means the result is economically important.”
6. “A large biased sample can still give a misleading confidence interval.”

Part B - Python Applications

Exercise 17 - Confidence Interval for Apple Mean Daily Return

Use the file:

`data/Lecture 07 aapl daily returns 2012 2021.csv`

The relevant column is `daily_return`.

1. Load the dataset.
2. Compute the sample size, sample mean, sample standard deviation, and standard error.
3. Construct an approximate 95% confidence interval for the mean daily return.
4. As a simple reporting convention, annualize the point estimate and both interval bounds by multiplying by 252. Note that this is a linear rescaling of the mean and its interval; it ignores compounding, and it is not how one would annualize volatility.
5. Write a careful interpretation.

Exercise 18 - Confidence Level Sensitivity

Using the Apple daily returns from Exercise 17:

1. Construct 90%, 95%, and 99% confidence intervals for the mean daily return.

2. Put the results in one table.
3. Compare the interval widths.
4. Explain the precision-confidence tradeoff.

Exercise 19 - Compare Apple and Netflix Precision

Use:

```
data/Lecture 07 aapl daily returns 2012 2021.csv
```

and

```
data/Lecture 07 nflx daily log returns 2015 2026.csv
```

Note that the two series use different return definitions (simple returns for Apple, log returns for Netflix). For that reason, treat this exercise as a comparison of *estimation precision*, not as a ranking of which asset “earns more.”

1. Compute the mean, standard deviation, standard error, and approximate 95% confidence interval for each return series.
2. Which asset has the larger standard error?
3. Which asset has the narrower confidence interval for its mean?
4. Explain how both volatility and sample size affect precision.

Exercise 20 - Confidence Interval for Mean Loan Amount

Use the file:

```
data/Lecture 07 german credit.csv
```

The relevant column is the loan amount.

1. Load the dataset.
2. Compute the sample size, sample mean, sample standard deviation, and standard error of the mean loan amount.
3. Construct an approximate 95% confidence interval for the mean loan amount.
4. Write a careful interpretation, stating clearly what population or process the interval describes.
5. Explain why a confidence interval for a mean loan amount answers a different question from simply reporting the average loan amount in the sample.

Exercise 21 - Careful Precision Memo

Using results from Exercises 17-20, write a short memo that includes:

1. one point estimate;
2. one standard error;
3. one confidence interval;
4. one sentence distinguishing standard deviation from standard error;
5. one caution about interpreting the interval.